

Lesson 17

Topic: Working with Maps & Geocoding, Formatting & Annotations

Prerequisites: Download sales_with_geodata.csv file

Part 1: Maps & Geocoding

- **Task 1: Create a Basic Map Showing Sales by Country**

Step-by-Step Instructions

1. Import the CSV

- Open **Power BI Desktop**.
- Go to **Home > Get Data > Text/CSV**.
- Load the sales_with_geodata.csv file.

2. Check Data Fields

Make sure your dataset has at least:

- Country
- Sales
- Optional: Latitude & Longitude

3. Create a Map Visual

- In **Visualizations**, click on the **Map** visual (the globe icon).

4. Set the Fields

- **Location** → Drag Country
- **Size** → Drag Sales

Power BI uses Bing Maps for geocoding. It will automatically locate countries.

5. Optional: Use Latitude & Longitude for Accuracy

If available:

- **Latitude** → Drag Latitude field
- **Longitude** → Drag Longitude field
- **Location** field becomes optional when coordinates are used

Troubleshooting Geocoding

- If countries are not recognized:
 - Go to the **Data view**
 - Select the Country column
 - In the ribbon, set **Data Category** to **Country/Region**

Optional Formatting

- Use **Format pane** to customize:
 - **Data colors** for better contrast
 - **Map styles** (road, aerial)
 - Enable or disable **Zoom controls, Bubbles, Labels**

- **Task 2: Use Latitude and Longitude for Precise Mapping**

Step-by-Step Guide

1. Make Sure Your Data Has Coordinates

Confirm your sales_with_geodata.csv contains these columns:

- Sales
- Latitude
- Longitude
- (Optional: City, Country)

2. Load Data

- Go to Home > Get Data > Text/CSV → select sales_with_geodata.csv.
- Click Load.

3. Insert a Map Visual

- In the Visualizations pane, select the regular Map visual (not Filled Map or Azure Map).
- Drag it onto the canvas.

4. Assign Fields

- Latitude → drag your Latitude column

- Longitude → drag your Longitude column
- Size → drag Sales to show bubble size based on sales value
- (Optional) Tooltips → drag fields like City, Country, or Sales for hover details

Power BI Note:

Power BI requires both Latitude and Longitude fields for precise point placement — don't use Location field if you're already using coordinates.

Optional Enhancements

- In Format pane:
 - Bubbles → adjust size, color, transparency
 - Map style → change to Aerial, Dark, etc.
 - Zoom → turn off auto-zoom for better control

• Task 3: Plot Total Sales by City

Step-by-Step Instructions

1. Check Your Data

Ensure your dataset includes:

- City
- Sales
- (Optional: Country, Latitude, Longitude)

2. Insert a Map Visual

- Open the **Visualizations** pane
- Select the **Map** visual (globe icon)

3. Drag Fields into the Map Visual

- **Location** → Drag City
- **Size** → Drag Sales

- (Optional) **Tooltips** → Drag Country, Sales, or any other descriptive info
Power BI will **automatically aggregate** sales by each city and **geocode** the location using Bing Maps.

If Cities Don't Display Correctly:

To improve location accuracy:

1. Go to **Data View**
2. Select the City column
3. In the ribbon, go to **Column Tools > Data Category** → set it to **City**

If still ambiguous, **combine City and Country** into a new column like:

powerquery

Location = City & ", " & Country

Then use that Location column as the **Map Location**.

Optional Formatting

- In the **Format** pane:
 - Adjust **bubble size**
 - Change **color** based on sales
 - Enable **category labels** to show city names

Result:

Your map now shows **bubbles representing total sales per city**. Larger bubbles = higher sales.

- **Task 4: Create a Region-wise Sales Heatmap (Use the Shape Map visual)**

Enable Shape Map Visual

If it's not visible:

- Go to **File > Options and settings > Options**
- Under **Preview Features**, check  **Shape Map Visual**

- Restart Power BI Desktop

2. Check Your Data

You should have:

- Region or State or Country (e.g., "California", "Texas")
- Sales

Power BI Shape Map supports built-in geographic keys like **US States**, **World countries**, **Counties**, etc.

3. Insert Shape Map

- In **Visualizations**, select the **Shape Map** icon (looks like a polygon)
- Add it to your report canvas

4. Assign Fields

- **Location** → Drag your Region or State field
- **Color saturation** → Drag Sales

This creates a **heatmap** where **darker colors = higher sales**

5. Fix Region Mapping (if needed)

If the map doesn't show properly:

- Go to the **Data View**
- Select your Region or State column
- In the ribbon: **Column Tools > Data Category** → set to **State or Province**, or **Country/Region**, depending on your data

6. Optional: Use Custom Map Files (.TopoJSON)

To use **custom regions** (like districts or business zones), upload a .topojson file:

- In the Shape Map visual, go to **Format > Shape**
- Enable "Map" → click + **Add Map** → upload .topojson file

Examples: <https://github.com/deldersveld/topojson>

Customize the Heatmap

In the **Format** pane:

- Turn on **Data Labels** to show values

- Change **Color scale** to better represent low vs high sales
- Adjust **Zoom, Pan, and Legend**
- **Task 5: Add Map Drill-Down from Country → Region → City**

Step-by-Step Instructions

1. Prepare Your Data

Ensure your dataset has these fields:

- Country
- Region or State/Province
- City
- Sales

Each field should be **categorized** properly (see Step 6).

2. Create a Location Hierarchy

1. In the **Fields** pane:
 - Right-click on the Country field → **New hierarchy**
 - Drag Region and City into that hierarchy

Now you should have a hierarchy like:

mathematica

Country Hierarchy

├── Country

├── Region

└── City

3. Insert a Map Visual

- Choose the regular **Map** visual (not Shape Map or Azure Maps).
- Add it to the canvas.

4. Assign Fields

- **Location** → Drag the **Country Hierarchy** into the Location bucket

- **Size** → Drag Sales
- (Optional) **Tooltips** → Add Sales, Country, Cit

5. Enable Drill-Down

At the top-right of the visual, activate the drill controls:

- **Click the Drill Icon** (looks like a pitchfork or arrow)
- Use **“Go to the next level”** to step through each level

Enable “Drill mode” to drill into a specific area (clicking a country drills into its regions)

6. Ensure Correct Geocoding

For precise mapping:

- Go to **Data view**
- Select:
 - Country → Set **Data Category** to **Country/Region**
 - Region → Set to **State or Province**
 - City → Set to **City**

This ensures Power BI geocodes each level correctly.

Result:

You now have an **interactive map with drill-down**:

- Default shows **sales by country**
- Click a country to drill into **regions**
- Click again to drill into **cities**

Optional Enhancements:

- Add **Slicers** for filtering by product or date
- Use **Bookmarks** to switch between map views
- Use **Azure Maps** for enhanced visuals (road, terrain, satellite)

Part 2: Formatting & Annotations

- **Task 6: Customize Map Visual – Titles, Colors, and Labels**

Select Your Map Visual

Click on the **Map** visual you created earlier (basic map, shape map, or Azure Map).

2. Customize the Title

- Go to the **Format pane** (paint roller icon)
- Expand the **Title** section
- Toggle it **On**
- Edit:
 - **Title Text** → e.g., "Sales by Country and City"
 - **Font family** → e.g., Segoe UI
 - **Font size** → e.g., 16–20 pt
 - **Alignment** → Center or Left
 - **Color** → Choose a contrasting color

3. Change Bubble or Region Colors

Based on the visual you're using:

For Map (bubble map):

- In the **Format pane**, go to:
 - **Data colors** → Choose a gradient or specific color for sales
 - Use **conditional formatting** (based on value)

For Shape Map:

- Go to **Color scales**
 - Set **Minimum – Maximum** color range
 - Choose colors like:
 - Low: Light yellow
 - High: Dark red
 - Or use **diverging** colors (green → yellow → red)

For Azure Map:

- Choose **Themes** like Road, Dark, Grayscale under **Map Styles**

4. Turn on Labels (Category or Data)

For Bubble Map:

- Go to **Format > Category Labels**
 - Toggle **On**
 - It will show City or Region name beside each bubble

For Shape Map:

- Toggle **Data labels** → On
 - You can show sales values inside each region

5. Adjust Bubble Size

- Go to **Bubbles** section in Format pane
 - Adjust **Min/Max size** if bubbles are too small or too large

6. Enable Zoom & Pan

- Go to **Map controls**
 - Turn **Zoom buttons** → On
 - Toggle **Auto zoom** → On/Off depending on user control

- **Task 7: Use Conditional Formatting for Bubble Colors**

For a Bubble Map:

1. Click on your map visual.
2. Go to the **Format pane > Data colors**
3. Click on the **fx** button next to the color
4. Choose:
 - **Based on field** → Sales
 - **Summarization** → Sum (or as needed)
 - **Color scale** → Choose a diverging or gradient color palette
(e.g., *light blue = low sales* → *dark blue = high sales*)

For a Shape Map:

- Go to **Color scales** under Format pane.
- Set **Minimum**, **Center** (optional), and **Maximum** values.
- Choose appropriate colors (e.g., Green-Yellow-Red).

- **Task 8: Add Annotations to Highlight Key Cities**

Option 1: Smart Tooltips

- Go to **Tooltips** bucket
- Add fields like:
 - City
 - Sales
 - Rank, or Special Notes (if you have a column for that)

Option 2: Add Text Boxes

- Go to **Insert > Text box**
- Add a floating label like:
 - “New York: Top Sales City – \$3.2M”
- Position it near the map corner

Option 3: Use Card Visuals + Filters

- Add a **Card** beside the map
- Use a **filter** or **Top N** logic to dynamically show "Top City" info

- **Task 9: Create a Tooltip Page with Detailed Info**

Step-by-step:

1. **Create a new page** → Rename it to City Tooltip
2. Go to **Page Information** (Format pane) → Turn **Tooltip** = On
3. Set **Page size** to **Tooltip** under Canvas Settings
4. Add visuals like:

- Table with all details for the selected city
- KPIs: Total Sales, Rank, Region
- Charts: Trend or pie chart

5. On your **Map visual**, do this:

- Format pane → **Tooltip** section
- Set **Type = Report page**
- Choose **City Tooltip**

- **Task 10: Add a Slicer to Filter the Map by Category**

Assuming your dataset has a column like Category (e.g., Product Category, Channel, Sales Type):

1. From **Visualizations pane**, select **Slicer**
2. Drag the Category field into it
3. Format the slicer:
 - Set to **Dropdown, List, or Buttons**
 - Add **search bar** for easier use
4. Your **map will now respond** to slicer selections